

Homework 4

Deadline: September 27

All sequences are real-valued.

Problem 1.

Suppose a sequence (s_n) satisfies

$$|s_{n+1} - s_n| \leq \frac{1}{n 2^n} \quad (n \in \mathbb{N}).$$

Prove that (s_n) is Cauchy and therefore has a limit $L \in \mathbb{R}$. Then prove that, for every $n \in \mathbb{N}$,

$$|s_n - L| \leq \frac{1}{n 2^{n-1}}.$$

Use finite geometric sums in your Cauchy argument, not a convergence test for infinite series.

Problem 2.

Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and suppose $\alpha = \sup A$ does not belong to A . Construct a *strictly increasing* sequence (a_n) of points of A such that

$$0 < \alpha - a_n < 2^{-n} \quad (n \in \mathbb{N}).$$

Justify both the construction and the conclusion $a_n \rightarrow \alpha$.

Problem 3.

Define

$$x_1 = 4, \quad x_{n+1} = \frac{3x_n + 4}{x_n + 3} \quad (n \geq 1).$$

Prove that the recurrence is well defined, that $2 < x_n \leq 4$ for every n , and that (x_n) is strictly decreasing. Deduce that (x_n) converges and determine its limit. Establish convergence before taking limits in the recurrence.

Problem 4.

Let

$$a_n = 2 + (-1)^n + \frac{1 + (-1)^n}{n} \quad (n \in \mathbb{N}).$$

Exhibit a constant subsequence and a convergent, nonconstant subsequence with a different limit. Use these subsequences to decide whether (a_n) converges, and justify your conclusion.

Problem 5.

Let $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ be strictly increasing, and suppose that $s_{\sigma(k)} \rightarrow L \in \mathbb{R}$. Construct a strictly increasing map $\tau : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$|s_{\sigma(\tau(j))} - L| < 2^{-j} \quad (j \in \mathbb{N}).$$

Prove explicitly that $\sigma \circ \tau$ is strictly increasing. Conclude that $(s_{\sigma(\tau(j))})$ is a subsequence of the original sequence (s_n) and converges to L .